

Assignment F

Common Notations

We use the following notation:

S_t	:	Stock price at time t
σ	:	Volatility of the Stock (assumed constant)
r	:	Interest rate
T	:	Time to option expiration (in years)
K	:	Strike price
W_t	:	Brownian motion process
$N(0, 1)$	:	Standard normal distribution
$(A - B)^+$	:	$\text{Max}(A-B, 0)$

Assignment F (Python)

- ▶ Due: April 7 by 2 PM.
- ▶ Write a function to price European Call options using Black Scholes formula.
- ▶ Measure time taken to price 1 million options.